

Indian government starts work on 5G



The Indian government it seems is aiming to adopt 5G technology alongside other developed markets, given its focus on programmes like Digital India. The government has commissioned a research team on 5G technology, which

has already filed 100 patents so far, of which around 10 have already been granted. This team comprises researchers from IISc Bangalore, IIT Bombay, IIT Hyderabad, IIT Madras and Centre of Excellence in Wireless Technology, IIT Madras.

Kiran Kuchi, professor at department of electrical engineering at IIT Hyderabad has said, "We are scaling up fast and switching gears because 5G is just around the corner." To aid these researchers, companies like Reliance Jio, Tejas Networks and other similar startups have also offered help by lending their manpower and equipment.

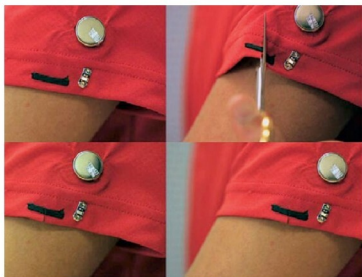
5G is the fifth generation of wireless networks that are expected to offer not just more speed for end users, but also have the capacity to connect billions of devices like driverless cars and household appliances that will connect to the Internet in the near future.

Magnetic ink that lets gadgets self-heal

A research by the engineering lab team at University of California, San Diego, USA, is hoping to fix gadgets by creating magnetic ink particles that could self-heal these devices when these break. Sensors printed with this ink would magnetically attach to each other when a rip or tear occurs, automatically fixing the device at the first sign of disintegration.

The self-healing effect is brought upon by the use of pulverised neodymium magnets in the ink, most often found in refrigerators and hard drives. This process is a lot faster than the conventional method of heating chemicals, which would take up to hours a day.

Amay Bandodkar, a member of the team, has compared the ink's self-healing ability to that of our skin. He has estimated that about US\$ 10 worth of ink can create hundreds of small devices. This will help in waste reduction, because the need for throwing broken gadgets will reduce after the self-healing ink comes into action.



The magnetic ink team is evaluating the best ink ratios for different gadget-printing applications, so as to create anything from solar panels to medical implants (Image courtesy: Science Advances)